Qually: An Accessible LLM Audit Tool for Transparent and Reflexive Qualitative Research

Kartik Trivedi
University of New Hampshire
kartik.trivedi@unh.edu

November 5, 2025

Abstract

Large Language Models (LLMs) are increasingly embedded in qualitative research workflows, yet most interfaces provide limited support for auditability, reproducibility, and reflexive practice. *Qually* is a lightweight, open tool that helps researchers design and run structured LLM experiments, capture prompt–response trails, and export analysis-ready artifacts. The tool supports multiple providers (OpenAI, Anthropic, Google, Mistral, xAI, DeepSeek), encrypted key storage, rate-limited API calls, and GUI-driven data import, prompt management, experiment setup, and results export. We describe the motivation for such a tool, its architecture and implementation, and a use case demonstrating codebook-constrained analysis with audit-ready JSONL exports and memo fields for reflexivity. We conclude with methodological implications for qualitative AI, limitations, and directions for community extension.

1 Introduction

LLMs enable rapid text generation, classification, and synthesis that can augment qualitative workflows. However, three challenges impede rigorous adoption: (i) insufficient **transparency** of prompt–response histories; (ii) limited **reproducibility** across models, parameters, and runs; and (iii) weak support for **reflexive practice** (e.g., documenting researcher stance, uncertainty, and constraints). Tooling that foregrounds these concerns is needed for trustworthiness in qualitative research [Lincoln and Guba, 1985, Braun and Clarke, 2006, Miles et al., 2014, Bender et al., 2021, Bommasani et al., 2021].

Contributions. This paper presents:

1. **Qually**, an accessible GUI and Python backend for audit-ready LLM experiments integrating encrypted API key management, rate-limited multi-provider access, and structured exports.
2. A **design** that aligns software affordances with qualitative methods principles: audit trails, memoing, and codebook-constrained procedures.
3. A **use case** showing how researchers can import a corpus, define prompts, and export JSONL/CSV artifacts suitable for downstream analysis and preregistration.

2 Related Work

LLMs and qualitative methods. The promise and risks of generative models in social science include scale and speed benefits alongside concerns about opacity, bias, and reproducibility [Bender et al., 2021, Bommasani et al., 2021]. Established qualitative standards emphasize trustworthiness, reflexivity, and systematic documentation [Lincoln and Guba, 1985,

Braun and Clarke, 2006, Miles et al., 2014, Charmaz, 2014]. Qually seeks to bridge these domains with pragmatic tooling.

Reproducibility and audit trails. In ML and HCI, open artifacts and experiment structure are central to reproducibility [Pineau et al., 2021]. In qualitative research, *audit trails* enable credibility and confirmability [Lincoln and Guba, 1985]. Qually operationalizes these principles via exportable experiment specifications, logs, and outputs tied to prompts, parameters, and model versions.

Existing tools. Commercial CAQDAS with AI features (e.g., NVivo, Atlas.ti) often restrict transparency into prompt flows and model configuration; research assistants (e.g., Elicit) favor discovery over auditability. Open-source options exist but rarely integrate multi-provider access, encryption, and GUI-based experiment management in one package. Qually aims to complement these ecosystems by focusing on auditability and method alignment.

3 Design Goals and System Overview

Qually is designed around four goals: (G1) **Auditability** of prompt–response trails; (G2) **Reproducibility** across providers, models, and parameters; (G3) **Accessibility** via a cross-platform GUI; and (G4) **Method alignment** with qualitative standards (e.g., memoing, codebooks).

3.1 High-Level Architecture

The core components include:

- **APIKeyManager**: encrypted storage and retrieval of provider keys (Fernet symmetric encryption).
- **PromptManager**: persistent prompt library with tags, optional prepend data, and system/user roles.
- **ExperimentManager**: structured experiment definitions (conditions = provider+model+parameters) and run orchestration with retry and rate-limiting.
- **GUI (PyQt)**: tabs for API keys, data import, prompts, experiments, and results; progress and logging widgets.

3.2 Data Flow

A typical session: (1) select a project folder; (2) add API keys for one or more providers; (3) import a CSV/JSON/Excel file and choose `id` and `text` columns; (4) author prompts (optionally with prepend text and tags); (5) define experiment conditions (provider, model, temperature, max tokens, etc.); (6) run experiments (threaded to keep the GUI responsive); and (7) export results (CSV/JSONL) and logs.

4 Implementation

Providers and parameters. Qually currently supports OpenAI, Anthropic, Google (Gemini), Mistral, xAI (Grok), and DeepSeek via REST APIs with configurable parameters (e.g., temperature, top-p, penalties, max tokens). Provider-specific defaults are chosen conservatively to reduce rate limits.

Encryption and key management. API keys are encrypted at rest using the Fernet scheme from the `cryptography` library. Keys are stored separately from experiment data, and the encryption key is persisted in a dedicated file to allow reuse across sessions.

Rate limiting and retries. The HTTP session is configured with exponential backoff and jitter; requests throttle based on provider-specific intervals to mitigate 429 and 5xx errors. All responses and errors are logged (with truncation where appropriate).

Data import and export. CSV/JSON/Excel imports include automatic column detection (e.g., `text`, `content`) with manual override. Exports include structured JSONL with `doc_id`, prompt metadata, model parameters, and raw text output, enabling downstream coding and analysis.

GUI. The PyQt interface wraps the backend managers, offering a tabbed workflow with contextual help, preview tables, and progress indicators. The design favors transparency (explicit parameter controls) and discoverability (tooltips, default prompts).

5 Use Case: Codebook-Constrained Thematic Coding with Audit Trail

We illustrate a common pattern: codebook-constrained coding for interview transcripts. The researcher loads a CSV with `id` and `text` columns, prepares a system prompt that constrains the model to an existing codebook, and runs multiple conditions (e.g., two providers $\times$ two temperatures).

5.1 Experiment Setup

Prompts can include prepend text with instructions such as: "Apply only the uploaded codebook; if none fit, use `UNSURE`." Researchers can also append reflexivity notes (e.g., stance, uncertainty) at run time for inclusion in exports.

5.2 Export Artifacts

Qually exports JSONL rows suitable for downstream validation or secondary analysis. Listing 1 shows a stylized example (abbreviated).

Listing 1: Abbreviated JSONL export for audit-ready analysis

```
{"doc_id":"INT01","prompt_id":"p001","provider":"openai","model":"gpt-4",
 "parameters":{"temperature":0.5,"top_p":1.0},
 "excerpt_span":[120,255],
 "codes":["ACCESS_BARRIERS","UNSURE"],
 "rationale":"Mentions platform friction; ambiguity about responsibility.",
 "confidence":0.72,
 "memo":"Reflexivity: I expect over-coding competence; checking for bias.",
 "output_text":"... final model response ..."}
```

5.3 Methodological Payoff

The export design supports inter-rater checks, preregistration of analysis plans, and reproducibility: identical prompts and parameters can be re-run across providers and versions to assess stability.

6 Evaluation

Our evaluation focuses on functionality and researcher experience rather than model accuracy. We validate (i) cross-provider runs with controlled parameters, (ii) integrity of audit exports, and (iii) responsiveness of the GUI under batch workloads. A formal user study and model performance benchmarking are out of scope but constitute future work.

7 Ethics, Privacy, and Data Governance

Researchers should avoid sending personally identifiable information (PII) to third-party APIs unless governed by IRB or equivalent agreements. Qually stores API keys encrypted and logs selectively; however, researchers remain responsible for data minimization, anonymization, and compliance with institutional policy.

8 Limitations and Future Work

Qually orchestrates provider APIs but does not itself validate the *truth* of outputs, guarantee fair use licensing, or eliminate model bias. Future work includes dataset-level caching, redaction tools, differential privacy options, and a headless CLI for scripted preregistrations and CI pipelines.

9 Availability

Source code, releases, and documentation will be hosted in a public repository with versioned archives (e.g., Zenodo DOI). We invite community contributions of new providers, evaluators, and example prompts.

Acknowledgments. We thank collaborators and early users for feedback on research affordances, audit needs, and GUI usability.

References

[Bender et al., 2021] Bender, E. M., Gebru, T., McMillan-Major, A., and Shmitchell, M. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pages 610–623.

[Bommasani et al., 2021] Bommasani, R., Hudson, D. A., Adeli, E., et al. (2021). On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*.

[Braun and Clarke, 2006] Braun, V. and Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2):77–101.

[Charmaz, 2014] Charmaz, K. (2014). *Constructing Grounded Theory*. SAGE, 2 edition.

[Lincoln and Guba, 1985] Lincoln, Y. S. and Guba, E. G. (1985). *Naturalistic Inquiry*. SAGE.

[Miles et al., 2014] Miles, M. B., Huberman, A. M., and Saldaña, J. (2014). *Qualitative Data Analysis: A Methods Sourcebook*. SAGE Publications, 3 edition.

[Pineau et al., 2021] Pineau, J., Vincent-Lamarre, P., Sinha, K., et al. (2021). Improving reproducibility in machine learning research. *Journal of Machine Learning Research*, 22(164):1–20.